

the extreme left-hand parabola (portion 0-1 of the graph). Please recall the formula for the distance covered in a uniformly accelerated motion with zero initial velocity.

Reader: This formula is $s(t) = \frac{at^2}{2}$ (1) where a is acceleration.

Author: And the extreme left-hand parabola is the graph of the function represented by your formula.

Reader: So for the time interval from 0 to t_1 the body moves at a constant acceleration.

Author: Exactly.

Reader: I see. For the time interval from t_1 to t_2 the body moves uniformly (portion 1-2 of the graph is a straight line); from t_2 to t_3 the body moves at a constant deceleration (the graph is an inverse parabola); from t_3 to t_4 the body is not moving at all; from t_4 to t_5 it moves at a constant acceleration; and from t_5 to t_6 it moves at a constant deceleration.

Author: Precisely so. Now let us consider the graph of the function $s(t)$ shown in Fig. 31 from a purely mathematical standpoint. Let us pose the following question: How strongly do the values of the function change in response to the value of its argument t in different portions of the graph?

Reader: In portion 3-4 the values of the function $s(t)$ do not change at all, while in other portions they do. A slower rate of change of the function is observed in the points ~~near~~ 0, 3, 4 and 6's vicinity; a faster rate of change is observed in the vicinity of points 1, 2 and 5. As a matter of fact, the rate of change is equally fast throughout portion 1-2.

Author: You are a keen observer. And where do you think the rate of change is faster, at point 2 or at point 5?

Reader: Of course, at point 2. Here the graph of the function has a much steeper slope than at point 5.